DRUGS · ANTIBIOTICS: OTHER CLASSES

Gentamicin (systemic)

Gentamicin is an aminoglycoside: rapidly bactericidal against Gram-negative bacilli, cheap, and synergistic with beta-lactams against streptococci and enterococci. It is also the most important cause of drug-induced deafness and vestibular failure worldwide. ENT surgeons meet it as a cause of ototoxicity, as a drug they must monitor in their own patients, as an ear drop, and as an intratympanic ablative treatment for Ménière's disease.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

GENTAMICIN (SYSTEMIC)

	SUMMARY
Class	Aminoglycoside (others: amikacin, tobramycin, streptomycin, neomycin)
How it works	Binds the 30S ribosomal subunit, causing misreading of mRNA; concentration-dependent, bactericidal, with a post-antibiotic effect
Spectrum	Enterobacterales, Pseudomonas, S. aureus (synergy); synergy with penicillins against streptococci and enterococci; no anaerobic activity; poor in acidic, anaerobic pus
ENT uses	Gram-negative sepsis; surgical prophylaxis in penicillin allergy (single dose with clindamycin); topical ear drops; intratympanic chemical labyrinthectomy
Adult dose	Once-daily dosing 5 to 7 mg/kg (based on ideal or adjusted body weight) with levels by a nomogram (for example the Hartford nomogram); divided dosing for endocarditis synergy (1 mg/kg every 12 hours)
Child dose	7 mg/kg once daily (neonates on specific regimens), guided by levels
Avoid	Myasthenia gravis; take care with renal impairment, pre-existing hearing loss, family history of aminoglycoside deafness, and other ototoxic or nephrotoxic drugs

	SUMMARY
Pregnancy	Avoid (fetal ototoxicity)

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

Uptake into bacteria needs oxygen-dependent transport, so aminoglycosides do not work against anaerobes or in abscesses. Once-daily dosing exploits **concentration-dependent killing** and the post-antibiotic effect while giving the kidney and ear a drug-free interval.

2.2 Handling

Not absorbed orally. Distributes in extracellular fluid, excreted unchanged by glomerular filtration; it accumulates in renal tubular cells and in the **perilymph and endolymph**, where it remains long after plasma levels fall.

3 CHAPTER THREE Ototoxicity

What the ENT surgeon must know.

3.1 How it damages the ear

Aminoglycosides enter hair cells through the mechanotransduction channels, generate reactive oxygen species and trigger apoptosis. **Gentamicin and streptomycin are mainly vestibulotoxic**; amikacin, kanamycin, neomycin and tobramycin are mainly cochleotoxic. Cochlear damage begins in the **outer hair cells**

of the basal turn, so high frequencies go first. Damage is usually **permanent** and can progress for weeks after the drug stops.

3.2 Who is at risk

- Duration over 5 to 7 days, high cumulative dose, high troughs, renal impairment and older age.
- Concurrent **loop diuretics**, vancomycin, cisplatin.
- **Mitochondrial 12S rRNA mutations** (m.1555A>G, m.1494C>T) cause profound deafness after even a single dose; in the UK, neonates are now tested rapidly for m.1555A>G before gentamicin where possible, and patients with a maternal family history of aminoglycoside deafness should not receive it.

3.3 Recognising vestibular toxicity

Bilateral vestibular loss presents with **oscillopsia** (the world jumps when walking) and imbalance in the dark, often noticed only when the patient is mobilised after an ICU stay; a head impulse test shows bilateral catch-up saccades. There is no vertigo, because both sides are lost together.

3.4 Monitoring

Measure levels by the protocol (a trough before the next dose with once-daily dosing), check renal function, question patients about hearing, tinnitus and imbalance, and consider high-frequency audiometry on long courses.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Ototoxicity** (cochlear and vestibular), as above.
- **Nephrotoxicity**: usually reversible acute tubular injury.

- Neuromuscular blockade, especially in myasthenia and with anaesthetic agents.

4.2 Interactions

Loop diuretics, vancomycin, ciclosporin, cisplatin, amphotericin and neuromuscular blockers.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Which aminoglycosides are **vestibulotoxic** versus **cochleotoxic**.
- Mechanism of hair cell damage and why it starts in the basal turn.
- The **m.1555A>G** mitochondrial mutation.
- How bilateral vestibular loss presents (oscillopsia) and how you test for it.
- Monitoring and risk factors.

5.2 Common pitfalls

COMMON PITFALLS

- Continuing gentamicin beyond a few days without levels and review.
- Giving it with furosemide without considering ototoxicity.
- Missing bilateral vestibular failure in a patient who was on ICU.

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